Midterm 2

Provided Reference Formulas:

Convolution and Transforms	$y[n] = \sum_k x[k]h[n-k]$ $\quad X(z) = \sum_n x[n]z^{-n}$ $X(e^{j\omega}) = \sum_n x[n]e^{-j\omega n}$ $\quad X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$
Frequency Mapping	$\omega = 2\pi\, f/f_s$ modulo 2π $\quad f_{alias} = \lvert f - kf_s \rvert$ in $[0, f_s/2]$
Quantization	$\Delta \approx$ full-scale range$/2^B$ $\quad SQNR_{FSsine} \approx 6.02B + 1.76$ dB
Filter Design	$h_d[n] = \frac{\sin(\omega_c n)}{\pi n},\ h_d[0] = \frac{\omega_c}{\pi}$ $h[n] = h[N-1-n], \quad H(e^{j\omega}) = e^{-jw(N-1)/2}A(\omega)$ $\tau_g(\omega) = -\frac{d}{d\omega}\angle H(e^{j\omega})$ $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$ $\quad z = \frac{1+sT/2}{1-sT/2}$ $\Omega = \frac{2}{T}\tan(\frac{\omega}{2})$ $\quad \omega = 2\tan^{-1}(\frac{\Omega T}{2})$